Week 11: Practical Issues in Research

Empirical Research Methods

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October 8, 2026

Learning Objectives

- Understand real-world challenges in implementing research
- Learn strategies for managing recruitment and attrition
- Understand compliance and spillover issues
- Learn about open science and pre-registration best practices

Challenges in Real-World Research

The design process

Randomize, measure outcomes, estimate effects. What is missing?

Answer: The messy reality of experimental execution and the pressure to publish

Recruitment is hard. Perfect populations are not always available. People drop out. Treatment doesn't always get delivered. Measurement is imperfect. Budgets are tight.

Do NOT cut corners

This is how you engage in questionable research practices (QRPs).

QRPs at UZH: Recruiting people to interact with AI on a platform that explicitly bans AI. Link Here.

Economics Example: Brian Wansink, economist whom we mentioned before, committed significant amounts of p-hacking. Link here

Recruitment Strategies

Additional examples in Axén et al. 2021,Treweek et al. 2018, Gkekas et al. 2023

Challenge

Finding enough participants who are willing to participate in your study.

Effective Strategies:

1. **Partner with organizations:** Work with schools, utilities, NGOs with existing relationships
2. **Leverage existing relationships:** Use facility managers, community leaders
3. **Offer incentives:** Financial (cash, gift cards) or non-financial (information, recognition)
4. **Multiple channels:** Online, door-to-door, community events
5. **Simplify participation:** Make it easy to enroll and easy to support enrollment (i.e. for clinicians)

Guideline: Survey dropout rate in similar studies and get percentage more than needed (based on power calculations) to account for attrition.

Selection Bias in Recruitment

Selection Bias in Recruitment

Volunteers are often more motivated, educated, or interested than average. Results may not generalize. Barrett, D'Agostino McGowan, and Gerke 2026.

Sources of Selection Bias:

- **Self-Selection:** Motivated people are more likely to volunteer
- **Convenience:** Recruiting from easily accessible locations excludes hard-to-reach groups
- **Eligibility Criteria:** Overly restrictive criteria filter out important populations

Solution: Adjusting for variables impacting uptake

- Use DAG to identify variables of concern
- Compare participants to target population
- Discuss external validity limitations

Attrition and Missing Data

Attrition can highly imbalance data and be as high as 30% in the field (Hewitt et al. 2010).

Types of Attrition

- **Unit Attrition:** Entire participant leaves study
- **Item Attrition:** Participant stays but specific data missing
- **Differential Attrition:** Dropout rates differ between treatment and control

Minimizing Attrition (with examples Gillies et al. 2021, Gkekas et al. 2023):

- Minimize burden (keep surveys short, automate data collection)
- Maintain engagement (newsletters, interim feedback, offer incentives)
- Offer completion incentives (bonuses, raffles)
- Monitor and follow up (track response rates, reach out to at-risk participants)

Imperfect Compliance

Recall from the previous lecture!
Some units don't receive assigned treatment (non-compliance). Some control units get treatment (crossover).

Different Approaches:

- **Intention-to-Treat (ITT):** Analyze by original assignment. Preserves randomization but underestimates true effect.
- **Treatment-on-Treated (ToT):** Analyze effect on those who complied. Requires IV to be valid.
- **Encouragement Design:** Instead of forcing treatment, randomize an incentive (e.g., email nudge). Creates valid instrument for double regression. (From email, Huang 2018)

Spillovers and SUTVA

Barrett, D'Agostino McGowan, and Gerke 2026

SUTVA (Stable Unit Treatment Value Assumption)

The treatment status of one unit depends only on their own treatment assignment, not others' treatment.

Spillovers result in a violation of SUTVA

- **Behavioral spillovers:** Neighbors talk, share information
- **Physical spillovers:** Shared infrastructure affects neighbors

Potential impact on results: Control group changes in the same manner as the treatment; estimated effect is underestimated.

Solution: Structural limitations on sharing experimental information

- Cluster randomization
- Track shared activities/spaces
- Document spillover afterwards to estimate rate

Questionable Research Practices

Motivation from "publish or perish" can incentivise manipulating analysis to generate false positives, cutting corners on protocol, or other QRPs.

- Around 25% of results using some common methods show signs of p-hacking, with RCTs less effected (Brodeur, Cook, and Heyes 2018).
- High proportion of economists admit to QRPs (Necker 2014).

Solutions: (From Dahlgren 2022 and Science 2026)

- **Pre-Registration:** Publicly register study before implementation
- **Pre-Analysis Plan:** Document analysis plan before seeing data (i.e. on AEA Registry, OSF)
- **Registered Report:** Peer review added before data is collected with publish guarantee (subject to second peer review, not significance)
- **Transparency:** Report all specifications, not just significant ones (commonly included above)
- **Reproduction and Replication:** Ensure results can be reproduced (using literally the same workflow) and replicated (another team conducting the same experiment)

Transparency and Open Science

Open Science

A movement to make scientific findings more accessible to everyone and strengthen reliability of research (Science 2026)

Sharing results AND methods fosters more discussion on research findings and allows for clearer review and elimination of mistakes and QRPs

Examples:

- **Open Data and Code:** Share raw data and scripts for reproducibility.
- **Open-Source Software:** Easier accessibility to same tools used for methods
- **Increase Open Collaboration:** Open reviews, adversarial collaborations, and more team science components

Ethics in Research

The newest texts in the field include Josephson and Michler 2023 and Siddiquee 2024. Below is a short summary

Key Ethical Principles

1. **Informed Consent:** Participants understand what they're agreeing to
2. **Minimizing Harm:** Research shouldn't cause damage
3. **Privacy:** Protect participant confidentiality
4. **Fairness:** Benefits and burdens distributed fairly

Institutional Review Board (IRB): An institutional solution

- Required for research involving human subjects
- Reviews study design, consent forms, data protection
- Get approval before starting research

Best Practices

- **Data Entry:** Use validation rules (i.e. check possible ranges)
- **Backup:** Regular backups in multiple locations
- **Documentation:** Document all variables, coding decisions
- **Version Control:** Track changes to data and code
- **Access Control:** Restrict who can modify data

Tools:

- Git can be used for data, as well as documentation and other version controls, but you must check storage location and access policies
- OSF can also be used for data
- R and Python (both open-source have commenting systems for reproducible analysis, as well as STATA (proprietary but common)
- Many, many more

The Most Modern of Issues: Responsible LLM Use

Rise in LLM chatbots has introduced a new range of opportunities, challenges, and problems to research ethics and best practices. MIT Taxonomy of AI Risks (including those beyond academics) is a good starter resource (Slattery et al. 2026).

Common currently understood issues include:

- Hallucinated citations
- Tortured phrases (mean-squared error $\rightarrow$ mean-squared blunder) and grammar (Field 2022)
- Over-generalization of claims in paper (Peters and Chin-Yee 2025)
- Cultural bias on behavioural studies (Tao et al. 2024)

Practical Checklist for Rigorous Experiments

Part 1: Planning

1. **Ethical Approval:** Secure IRB approval and clear informed consent
2. **Power Calculation:** Use power calculations to target 80% power for MDE
3. **Randomization Strategy:** Pick appropriate unit and method to limit spillovers
4. **Recruitment Plan:** Develop strategy for finding and retaining participants
5. **Pre-Registration:** Pre-register hypotheses, variables, and analysis plan

Practical Checklist for Rigorous Experiments

Part 2: Implementation

6. **Monitor Attrition:** Track dropouts, compare attrition by group
7. **Check Balance:** Verify treatment and control are balanced on baseline characteristics
8. **ITT Analysis:** Use ITT as primary analysis; IV/LATE as secondary
9. **Robustness Checks:** Test sensitivity to specification choices
10. **Open Data:** Share data and code for reproducibility

Summary: Practical Issues in Research

Key Concepts

- Real-world research faces recruitment, attrition, compliance, and spillover challenges
- Anticipate problems and plan solutions in advance
- Pre-registration and open science improve credibility
- Ethics and data quality ensure research stands up to scrutiny

Citations and Resources I

- Axén, Iben et al. (Jan. 2021). "Recruiting in intervention studies: challenges and solutions". In: *BMJ Open* 11.1, e044702. ISSN: 2044-6055. DOI: 10.1136/bmjopen-2020-044702. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC7839899/ (visited on 05/11/2026).
- Barrett, Malcolm, Lucy D'Agostino McGowan, and Travis Gerke (Mar. 2026). *Causal Inference in R*. URL: https://www.r-causal.org/.
- Brodeur, Abel, Nikolai Cook, and Anthony Heyes (2018). "Methods Matter: P-Hacking and Causal Inference in Economics". en. In: *SSRN Electronic Journal*. ISSN: 1556-5068. DOI: 10.2139/ssrn.3249910. URL: https://www.ssrn.com/abstract=3249910 (visited on 05/12/2026).

Citations and Resources II


Dahlgren, Peter M. (May 2022). *4 Remedies to questionable research practices | How Scientists Lie*. en. URL: https://howscientistslie.com/remedies.html (visited on 05/12/2026).


Field, Matt (Jan. 2022). *"Bosom peril" is not "breast cancer": How weird computer-generated phrases help researchers find scientific publishing fraud*. en-US. URL: https://thebulletin.org/2022/01/bosom-peril-is-not-breast-cancer-how-weird-computer-generated-phrases-help-researchers-find-scientific-publishing-fraud/ (visited on 05/12/2026).

Citations and Resources III


Gillies, Katie et al. (2021). "Strategies to improve retention in randomised trials". en. In: *Cochrane Database of Systematic Reviews* 3. ISSN: 1465-1858. DOI: 10.1002/14651858.MR000032.pub3. URL: https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.MR000032.pub3/full?highlightAbstract=randomis%7Crandomised%7Cmethod%7Cmethods%7Ccontrol%7Cimprov%7Ctrial%7Ccontrolled%7Crandom%7Ctrials%7Crecruitment%7Cimprove%7Crecruit%7Cto%7Crandomized (visited on 05/11/2026).


Gkekas, Athanasios et al. (July 2023). "A systematic review of economic evaluations alongside studies within a trial (SWATs) for improving recruitment and retention in randomised controlled trials". EN. In: *Research Methods in Medicine & Health Sciences* 4.3, pp. 94–112. ISSN: 2632-0843. DOI: 10.1177/26320843221147838. URL: https://doi.org/10.1177/26320843221147838 (visited on 05/11/2026).

Citations and Resources IV


Hewitt, Catherine E. et al. (Nov. 2010). "Assessing the impact of attrition in randomized controlled trials". In: *Journal of Clinical Epidemiology* 63.11, pp. 1264–1270. ISSN: 0895-4356. DOI: 10.1016/j.jclinepi.2010.01.010. URL: https://www.sciencedirect.com/science/article/pii/S0895435610000442 (visited on 05/12/2026).


Huang, Francis L. (Jan. 2018). "Using Instrumental Variable Estimation to Evaluate Randomized Experiments with Imperfect Compliance". None. In: *Practical Assessment, Research, and Evaluation* 23.1. ISSN: 1531-7714. DOI: 10.7275/k0p6-yj16. URL: https://openpublishing.library.umass.edu/pare/article/id/1661/ (visited on 05/12/2026).

Citations and Resources V

- Josephson, Anna and Jeffery D. Michler (Sept. 2023). *Research Ethics in Applied Economics: A Practical Guide*. en. URL: https://www.routledge.com/Research-Ethics-in-Applied-Economics-A-Practical-Guide/Josephson-Michler/p/book/9780367457419 (visited on 05/12/2026).
- Necker, Sarah (Dec. 2014). "Scientific misbehavior in economics". In: *Research Policy* 43.10, pp. 1747–1759. ISSN: 0048-7333. DOI: 10.1016/j.respol.2014.05.002. URL: https://www.sciencedirect.com/science/article/pii/S0048733314000900 (visited on 05/12/2026).
- Peters, Uwe and Benjamin Chin-Yee (Apr. 2025). "Generalization bias in large language model summarization of scientific research". In: *Royal Society Open Science* 12.4, p. 241776. ISSN: 2054-5703. DOI: 10.1098/rsos.241776. URL: https://doi.org/10.1098/rsos.241776 (visited on 05/12/2026).

Citations and Resources VI

- Science, Center for Open (2026). *What is open science?* en. URL: https://www.cos.io/open-science (visited on 05/11/2026).
- Siddiquee, Muhammad Shahadat Hossain (2024). "Ethical Issues in Research Practices in Economics". en. In: *Ethics in Social Science Research: Current Insights and Practical Strategies*. Ed. by Otto Federico von Feigenblatt of Braemar and M. Rezaul Islam. Singapore: Springer Nature, pp. 131–140. ISBN: 978-981-97-9881-0. DOI: 10.1007/978-981-97-9881-0_10. URL: https://doi.org/10.1007/978-981-97-9881-0_10 (visited on 05/12/2026).
- Slattery, Peter et al. (May 2026). "The AI risk repository: A meta-review, database, and taxonomy of risks from artificial intelligence". In: *Patterns* 7.5. arXiv:2408.12622 [cs], p. 101517. ISSN: 26663899. DOI: 10.1016/j.patter.2026.101517. URL: http://arxiv.org/abs/2408.12622 (visited on 05/12/2026).

Citations and Resources VII

- Tao, Yan et al. (Sept. 2024). "Cultural bias and cultural alignment of large language models". In: *PNAS Nexus* 3.9, pgae346. ISSN: 2752-6542. DOI: 10.1093/pnasnexus/pgae346. URL: https://doi.org/10.1093/pnasnexus/pgae346 (visited on 05/12/2026).
- Treweek, Shaun et al. (Feb. 2018). "Strategies to improve recruitment to randomised trials". en. In: *Cochrane Database of Systematic Reviews* 2018.2. Ed. by Cochrane Methodology Review Group. ISSN: 14651858. DOI: 10.1002/14651858.MR000013.pub6. URL: http://doi.wiley.com/10.1002/14651858.MR000013.pub6 (visited on 05/11/2026).